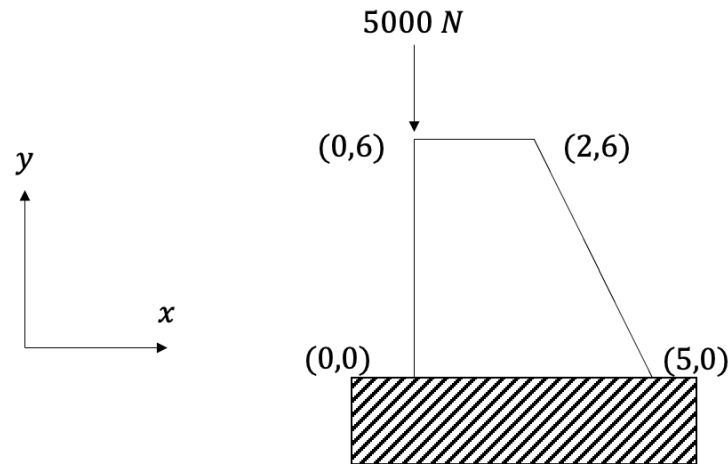


Instructions:

- Submit your assignment as a single PDF file through Canvas.
- Show all your steps and solution procedures.
- Make sure that your solution is well organised and that you are using appropriate headers for each question and sub-question.
- Scanned photos of your handwritten solution are acceptable as long as they are **legible**
- If a question requires plotting, please plot the expressions using any software/language (MATLAB, Excel, Mathematica, etc.) of your choice. However, **hand drawings** will not be accepted.
- You can use any software or calculator to perform any calculations. Using symbolic packages in MATLAB, Mathematica is acceptable as long as the solution steps are well laid out in your submission. **In case you use those packages please attach the script in your submission file as plain text and not a screenshot.**
- While you can discuss the problems with classmates, all submitted work must be your own to conform to academic integrity guidelines.
- Late assignments will not be accepted.

Question 1 (100 Points)

Consider the following structure. Solve for the nodal displacements using **1 linear quadrilateral element** assuming a plane stress condition (with unit thickness) and that the elastic modulus and Poisson's ratio are 20 GPa and 0.2 respectively.



- I. Determine the stiffness matrix of the element using a 2x2 Gauss Scheme. (50 Points)
- II. Solve for the nodal displacements using a 1x1 Gauss Integration scheme (Reduced Integration) (25 Points)
- III. Solve for the nodal displacements using a 2x2 Gauss Integration scheme (Full Integration) (25 Points)

NB: The coordinates are expressed in m.

Hint for Part II, the results may be surprising (read more about [spurious modes](#))